

Enhancing Belief Propagation Decoders for Correlated Errors Using Cycle Error Reconstruction

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Update to Steve

23 June 2026



Belief Propagation (BP) Decoders

- Linear time decoder for any (q)LDPC code for a stochastic error model : (p_x, p_z) .
- Classical BP algorithm independently on the Tanner Graphs describing X and Z stabilizers.

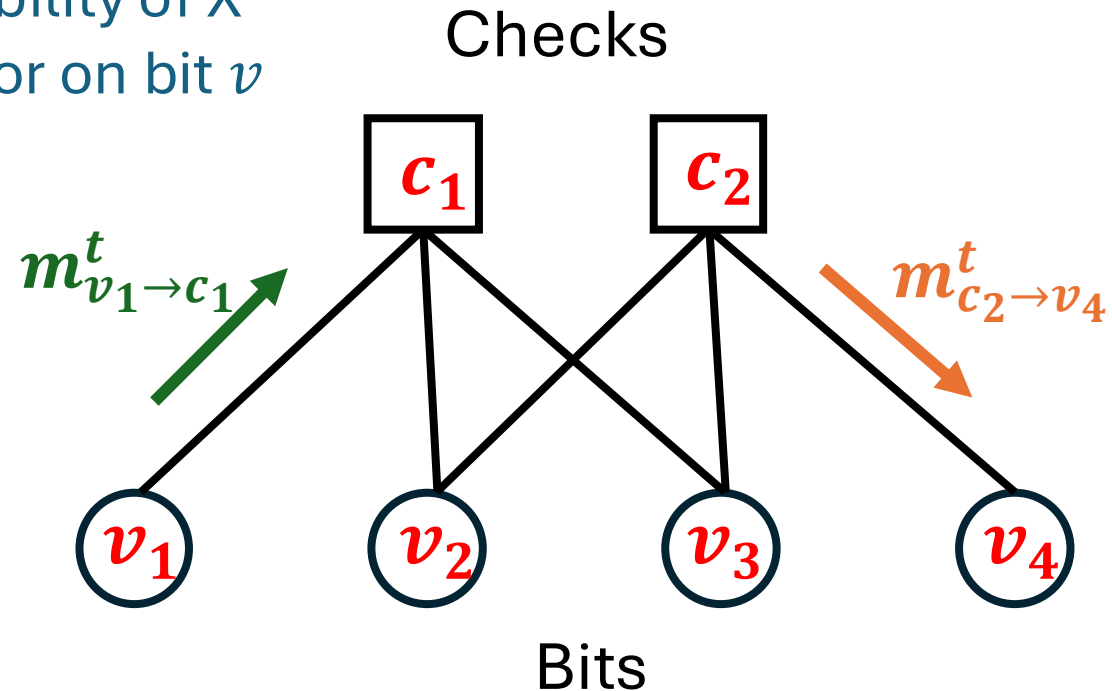
$$H = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{matrix} \mathbf{c}_1 \\ \mathbf{c}_2 \end{matrix}$$

$\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \mathbf{v}_4$

Log Likelihood Ratio = $l_v = \log \left(\frac{1-p_{x/z}}{p_{x/z}} \right)$ Probability of X (Z) error on bit v

$$m_{c \rightarrow v}^t = 2 (-1)^{s_c} \tanh^{-1} \left(\prod_{v' \in N(c)} \tanh \left(\frac{m_{v' \rightarrow c}^t}{2} \right) \right)$$

$$m_{v \rightarrow c}^{t+1} = l_v + \sum_{c' \in N(v) \setminus c} m_{v' \rightarrow c}^t$$



BP as a Neural Network

$$\text{Log Likelihood Ratio} = l_v = \log \left(\frac{1-p_{x/z}}{p_{x/z}} \right)$$

Repeat for T iterations

$$\log \tanh \left(\frac{m_{c \rightarrow v}^t}{2} \right) = i \pi s_c + \sum_{v' \in N(c) \setminus v} \log \tanh \left(\frac{m_{v' \rightarrow c}^t}{2} \right)$$
$$m_{v \rightarrow c}^{t+1} = l_v + \sum_{c' \in N(v) \setminus c} m_{v' \rightarrow c}^t$$

Compute refined beliefs

$$\mu_v^T = l_v + \sum_{c \in N(v)} m_{c \rightarrow v}^T$$

$\mu_v^T > 0$: no error

$\mu_v^T < 0$: error

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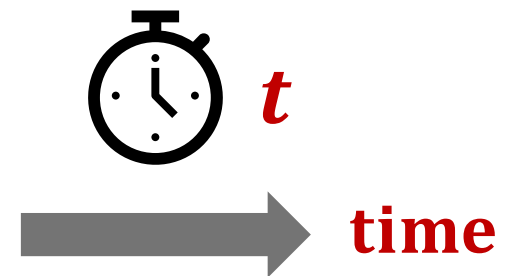
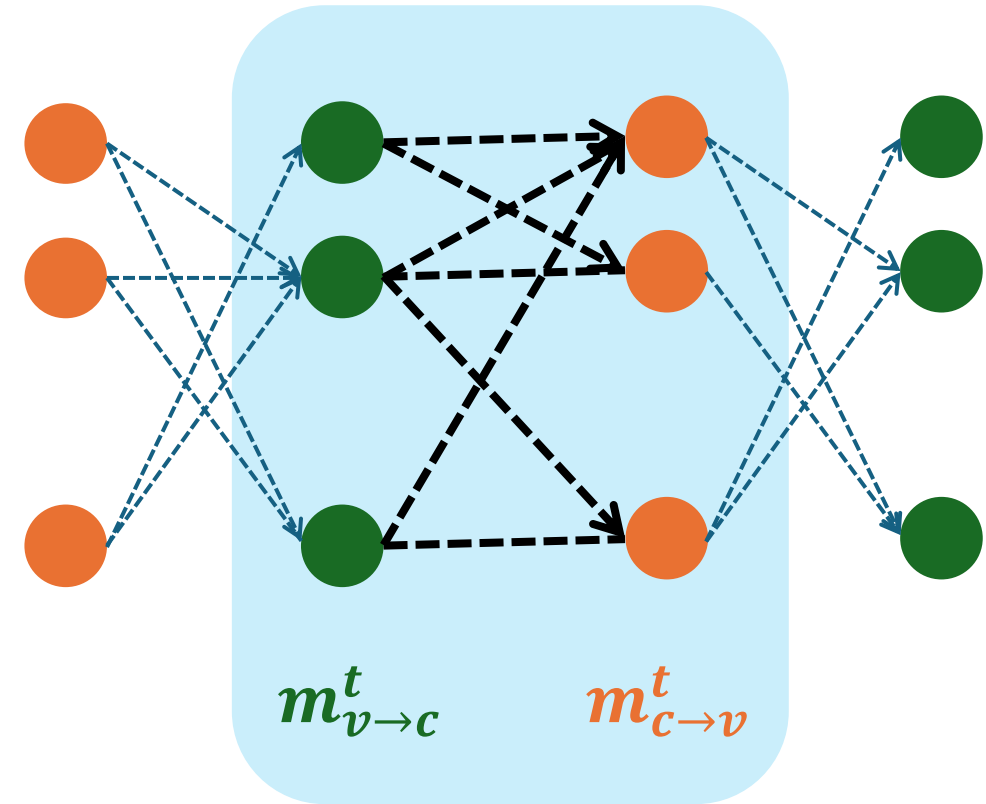
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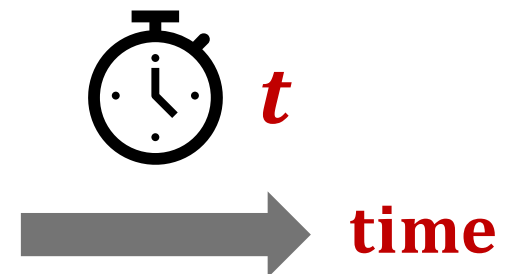
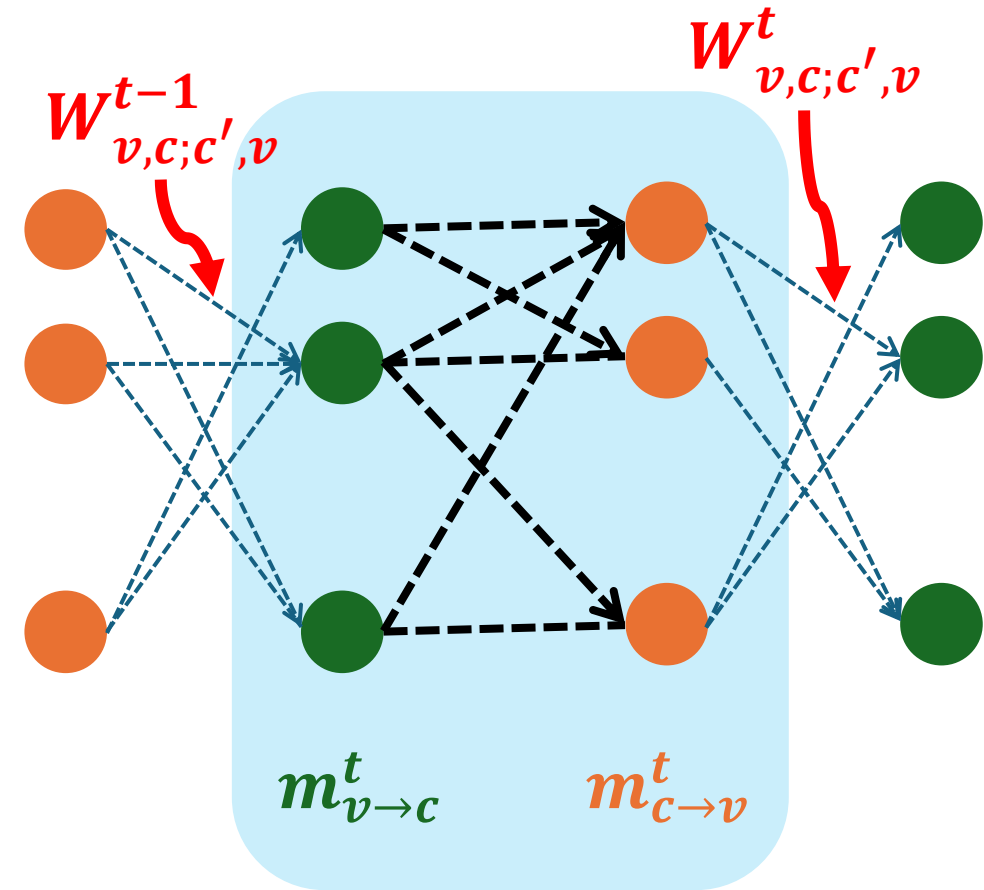
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$$m_{v \rightarrow c}^{t+1} = \mathbf{b}_v l_v + \sum_{c' \in N(v) \setminus c} \mathbf{W}_{v,c;c',v}^t m_{v' \rightarrow c}^t$$

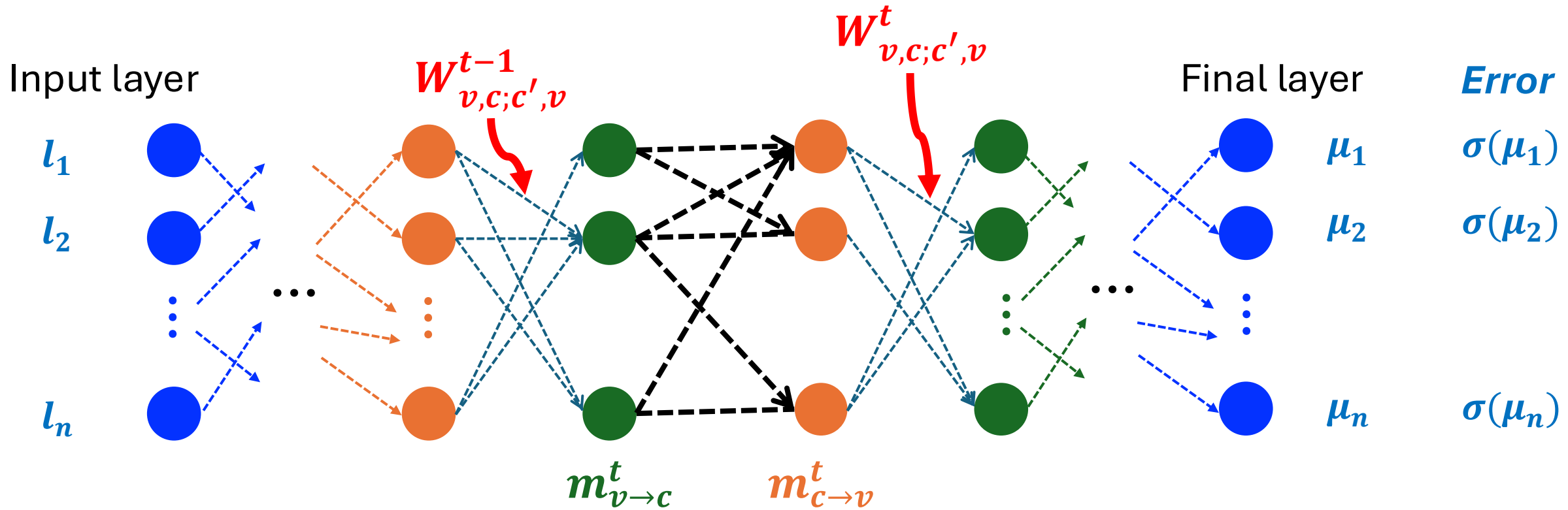
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BP as a Neural Network



Activation function at layers

$$a(x) = 2 \tanh^{-1}(e^x)$$

$$a(x) = \log \tanh(x/2)$$

Weights obtaining from training the NN:

$$s = H^\perp(\sigma(\mu_v) + e) \quad \text{Loss} = |\sin(\pi s_c/2)|.$$

Nachmani et. al., 2016. IEEE (Allerton)

Liu, Poulin, 2019. Phys. Rev. Lett. 122, 200501

Adding CER inputs to BP

Memory noise:

$$\mathcal{E}(\rho) = \sum_{i,j} \chi_{i,j} P_i \rho P_j$$

Pauli decay rates: $\chi_{i,i} \geq 0$

*Two-qubit correlations: spatial cross-talk,
frequency coupling (imperfect two-qubit gates)*

Single qubit error rates

$p_{i,I}, p_{i,X}, p_{i,Z}, p_{i,Y}$: on each qubit

Two-qubit error rates

$p_{ij,II}, p_{i,XX}, \dots, p_{ZZ}$: on qubit pairs

Weights in Neural BP: $W_{v,c;c',v}^t$ and b_v : minimize Loss per sample

$$\sum_c f\left(H_{\perp}(e + \sigma(\mu))_c\right) + \sum_{\langle i,j \rangle \in \mathcal{C}} \lambda_{i,j} \left(\sigma(\mu_i^T) - \sigma(\mu_j^T)\right)^2$$

Penalize configurations where error occurs on qubit i and no error on j .

Connectivity graph (hardware)

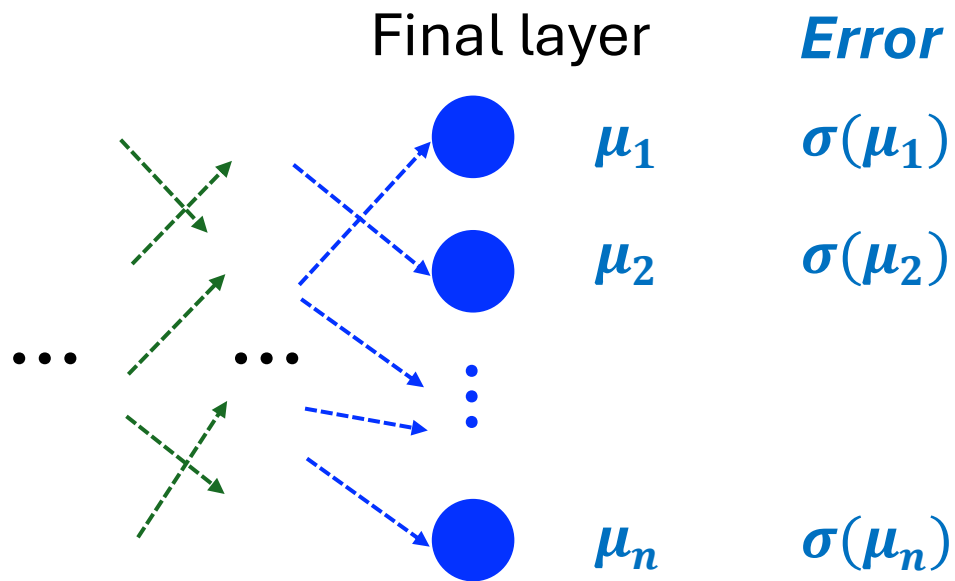
From CER

Fazio et al 2026. Phys. Rev. X 16, 011030

Harper and Flammia, 2023 . PRXQuantum.4.040311

Other additions to Loss

- We added two more considerations to the Loss function to encourage



$$\sigma(\mu_i) = \frac{1}{e^{\mu_i} + 1}$$

- Certainty**: Want confidence on the bit value, i.e. $|\mu_i| \gg 0$. Penalize LLRs near 0.

$$- \sum_{i=1}^n h(\sigma(\mu_i))$$

where $h(p)$: Binary entropy function

- Sparsity**: Prefer low weight solutions (more likely) before high-weight ones

$$+ \sum_{i=1}^n \sigma(\mu_i)$$

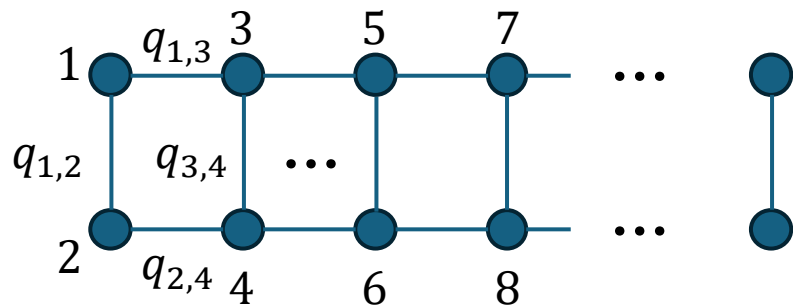
Likelihood of #1s in inferred error ($\mu_i \ll 0$) is minimized.

Numerical Results

Neural BP decoder vs. **BP-OSD decoder**
for correlated error models. (Monte Carlo)

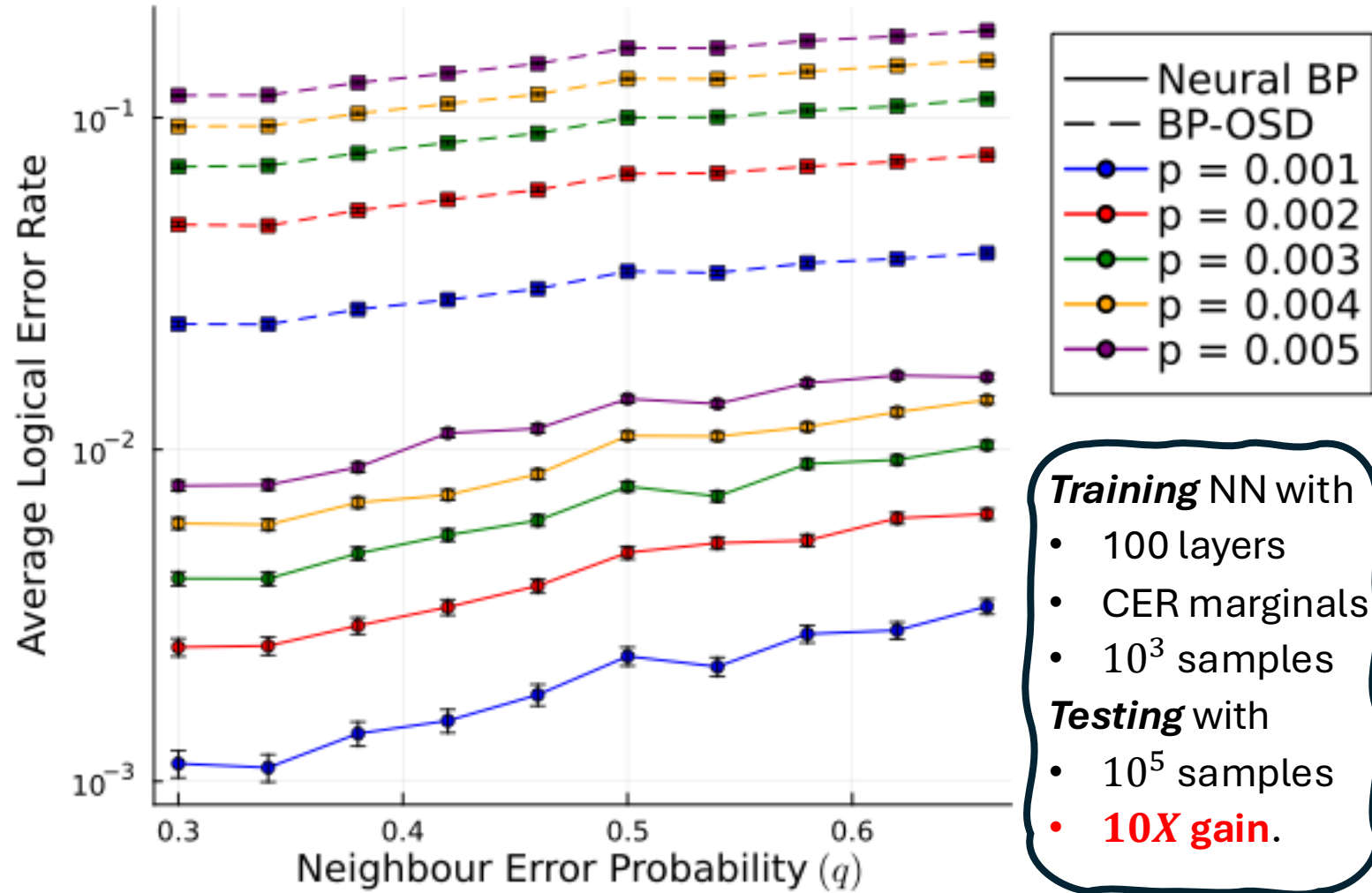
Error model: Ballistic Error Model

- Each qubit independently undergoes an error (X/Z) with probability p .
- Qubit i has an error $\Rightarrow$ each connected qubit j has error with probability $q_{i,j}$.



- CER provides two qubit marginals

Code: HGP of Hamming Codes: $[[58, 16, 9]]$



Nickerson and Brown, 2019, Quantum

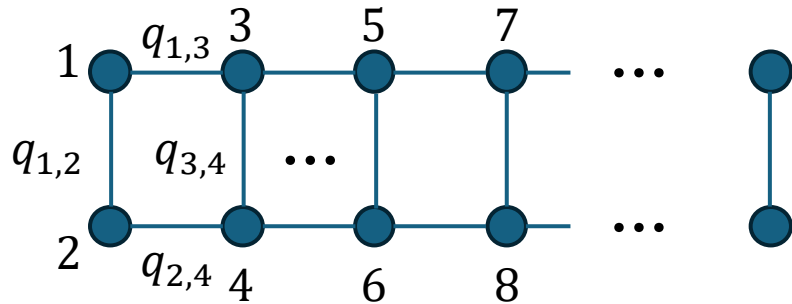
Roffe et. al. Phys. Rev. Research 2, 043423, 2020

Numerical Results

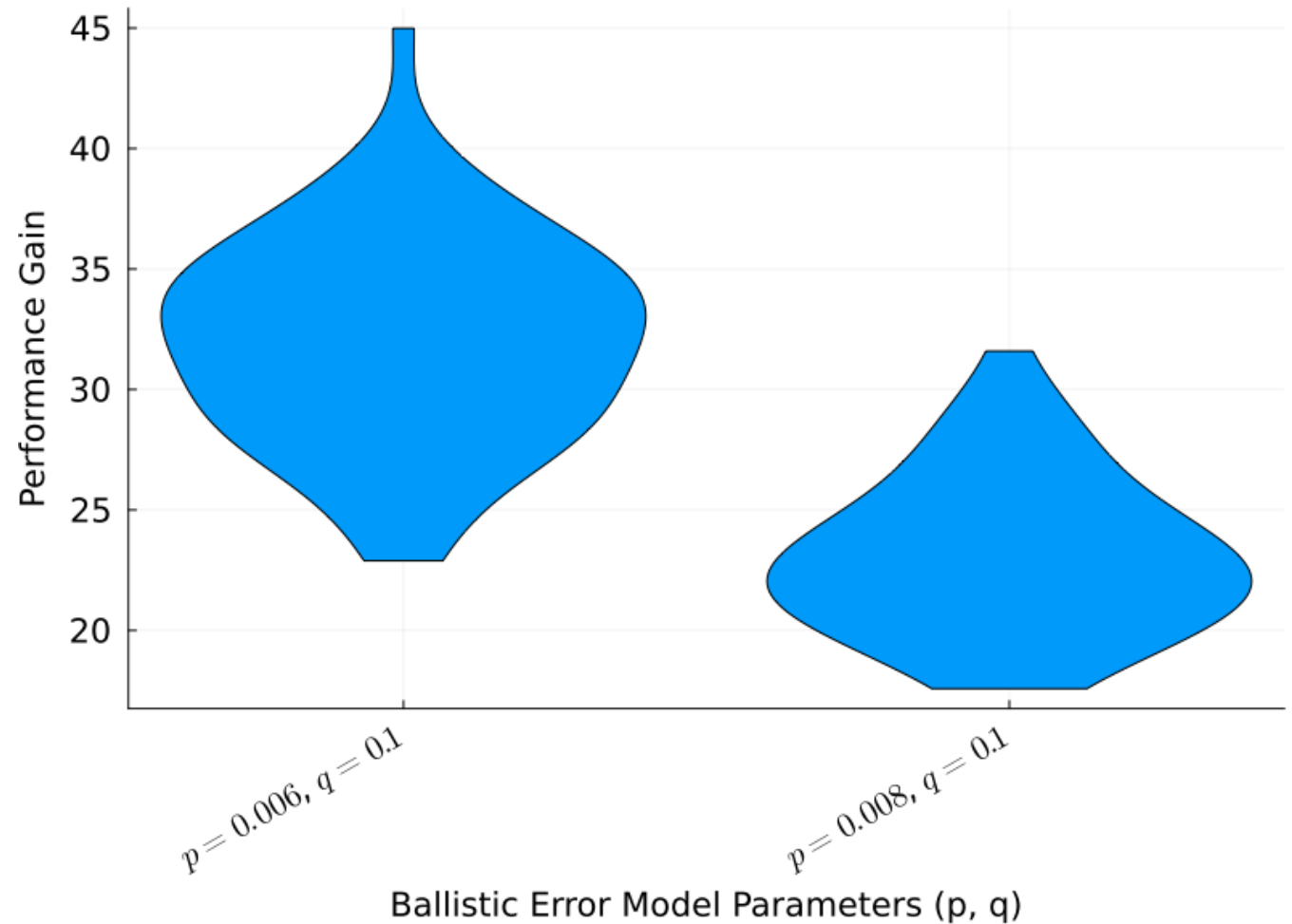
Neural BP decoder vs. **BP-OSD decoder** for Ballistic error model (gain for each sample)

Error model: Ballistic Error Model

- Each qubit error with probability p_i .
- Neighbouring qubit error with prob $q_{i,j}$.



- Sample p_i and $q_{i,j}$ from Gaussian $\mathcal{N}(q, 0.1)$.
- Bicycle Bivariate codes: $[[72,10,6]]$.

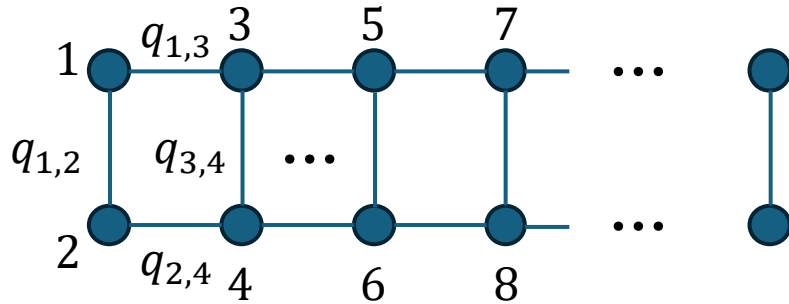


<https://www.nature.com/articles/s41586-024-07107-7>

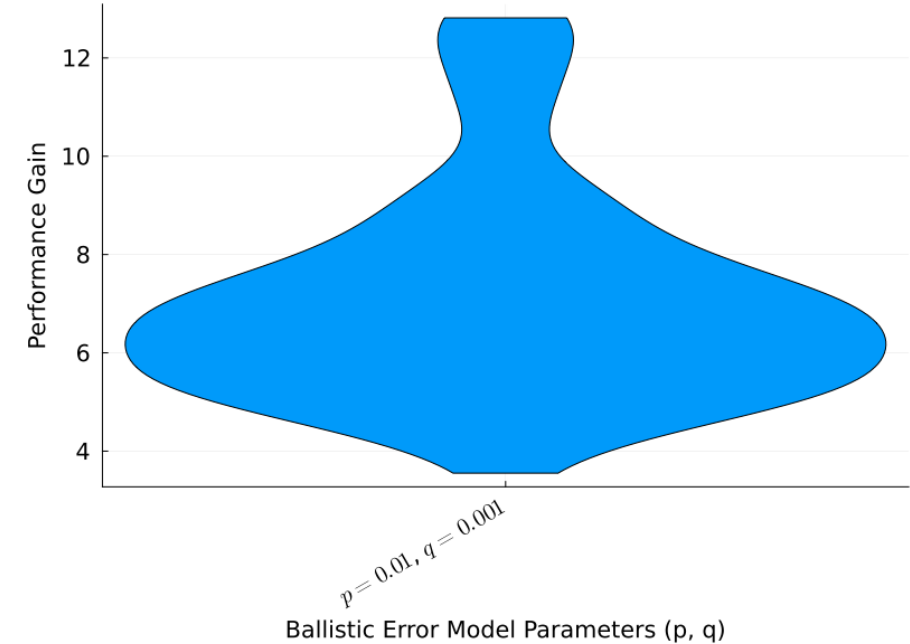
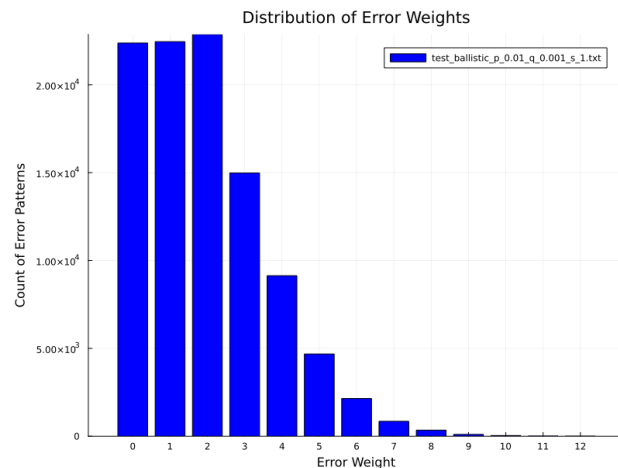
Typically ~30x gains over state of the art

Modelling cross-talk

- Single qubit errors occur with probability p_i and two qubit errors with $q_{i,j}$
- *Derived from an Ising model with transverse magnetic field (Debankan)*



$$p_i \sim N(\mu_p, \sigma_p), q_{i,j} \sim N(\mu_q, \sigma_q)$$



Small training instance: 100-layer Neural Network

Conclusions and future work

- What we have demonstrated performance gains over **BP-OSD** of:
 - 10X gain for Hypergraph Product Code under Ballistic Error Model Noise.
 - 30X gain for 72-qubit Bivariate Bicycle (BB) code under Ballistic Error Model.
 - 10X gain for 72-qubit Bivariate Bicycle (BB) code under Cross-Talk model.

<https://github.com/paviudes/CorrelatedBPDecoderWithCER>

- We are currently trying to demonstrate performance gains with:
 - A larger code in a family: 90-qubit BB code or 288 qubit HGP.
 - Trying to show numerical evidence for a threshold (better than BP-OSD)

Supplementary Slides

Error Model starting from Hamiltonian

$$H_{\text{err}} = \sum_i h_i z_i + \sum_{\langle ij \rangle \in E} J_{ij} z_i z_j$$

$\downarrow$ local dephasing $\downarrow$ residual ZZ cross-talk

$$\text{Error unitary } U_{\text{err}} = e^{-i H_{\text{err}} \tau} = \exp \left[-i \sum_i \theta_i z_i - i \sum_{\langle ij \rangle \in E} \phi_{ij} z_i z_j \right]$$

$$AU[z_i, z_{ij}] = 0 \rightarrow \text{commutes}$$

$$[z_i, z_j] = 0$$

$$\theta_i = h_i \tau$$

$$\phi_{ij} = J_{ij} \tau$$

$$U_{\text{err}} = \prod_i \exp(-i \theta_i z_i) \prod_{\langle ij \rangle \in E} \exp(-i \phi_{ij} z_i z_j)$$

$$= \prod_i \underbrace{(\cos \theta_i - i \sin \theta_i z_i)}_{U_1} \prod_{\langle ij \rangle \in E} \underbrace{(\cos \phi_{ij} - i \sin \phi_{ij} z_i z_j)}_{U_2}$$

$$U_1(\hat{P}) = (\cos \theta_i - i \sin \theta_i z_i) P (\cos \theta_i + i \sin \theta_i z_i)$$

$$= \cos^2 \theta_i P + \sin^2 \theta_i z_i P z_i + \underbrace{\text{off diagonal terms}}_{\substack{\downarrow \\ \text{Removed via RC}}}$$

$$= (1 - p_i) P + p_i z_i P z_i$$

$$U_2(\hat{P}) = \cos^2 \phi_{ij} P' + \sin^2 \phi_{ij} z_i z_j P' z_i z_j$$

$$= (1 - q_{ij}) P' + q_{ij} z_i z_j P' z_i z_j$$

$\therefore$ The effective error channel is

$$\mathcal{E} = \prod_i [(1 - p_i) I + p_i z_i] \prod_{\langle ij \rangle \in E} [(1 - q_{ij}) I + q_{ij} z_i z_j]$$

$\downarrow$ there exist Hamming wt > 2 terms also

Physically this means two error events happen independently.

Physical motivation

$p_i \sim \text{normal}(p_{\text{mean}}, \text{std})$
 $\rightarrow$ each qubit has different dephasing T_2 times

$q_{ij} \sim \text{normal}(q_{\text{mean}}, \text{std})$
 $\rightarrow$ different neighboring qubits have different residual cross-talk strengths.

Hypergraph Product Codes

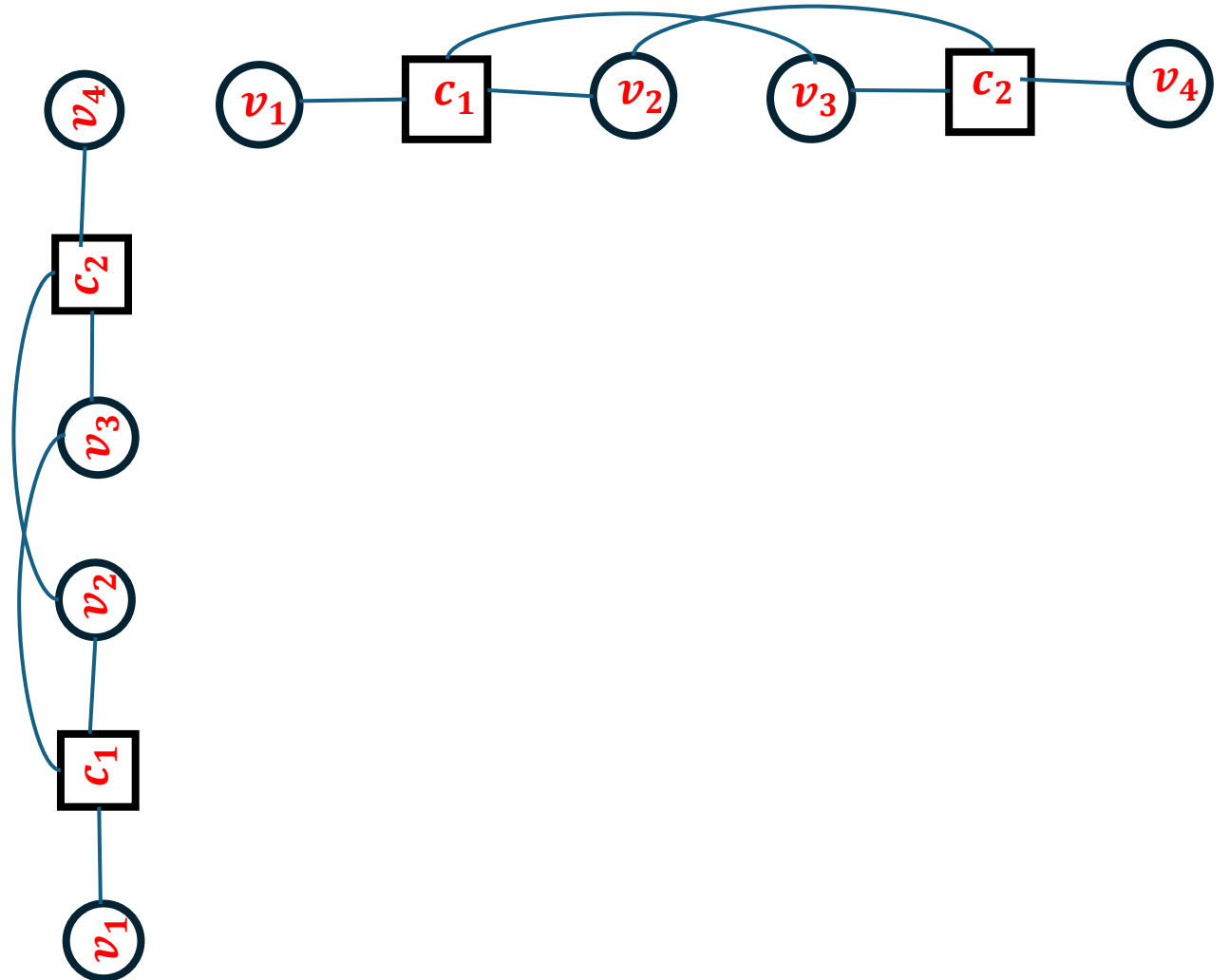
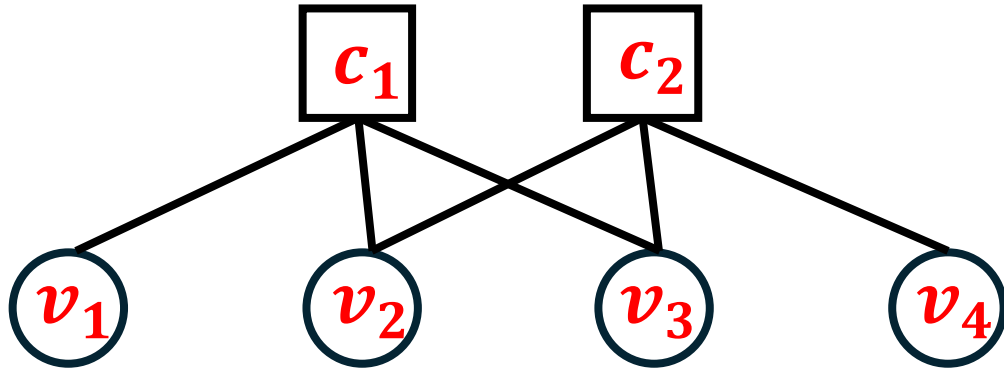
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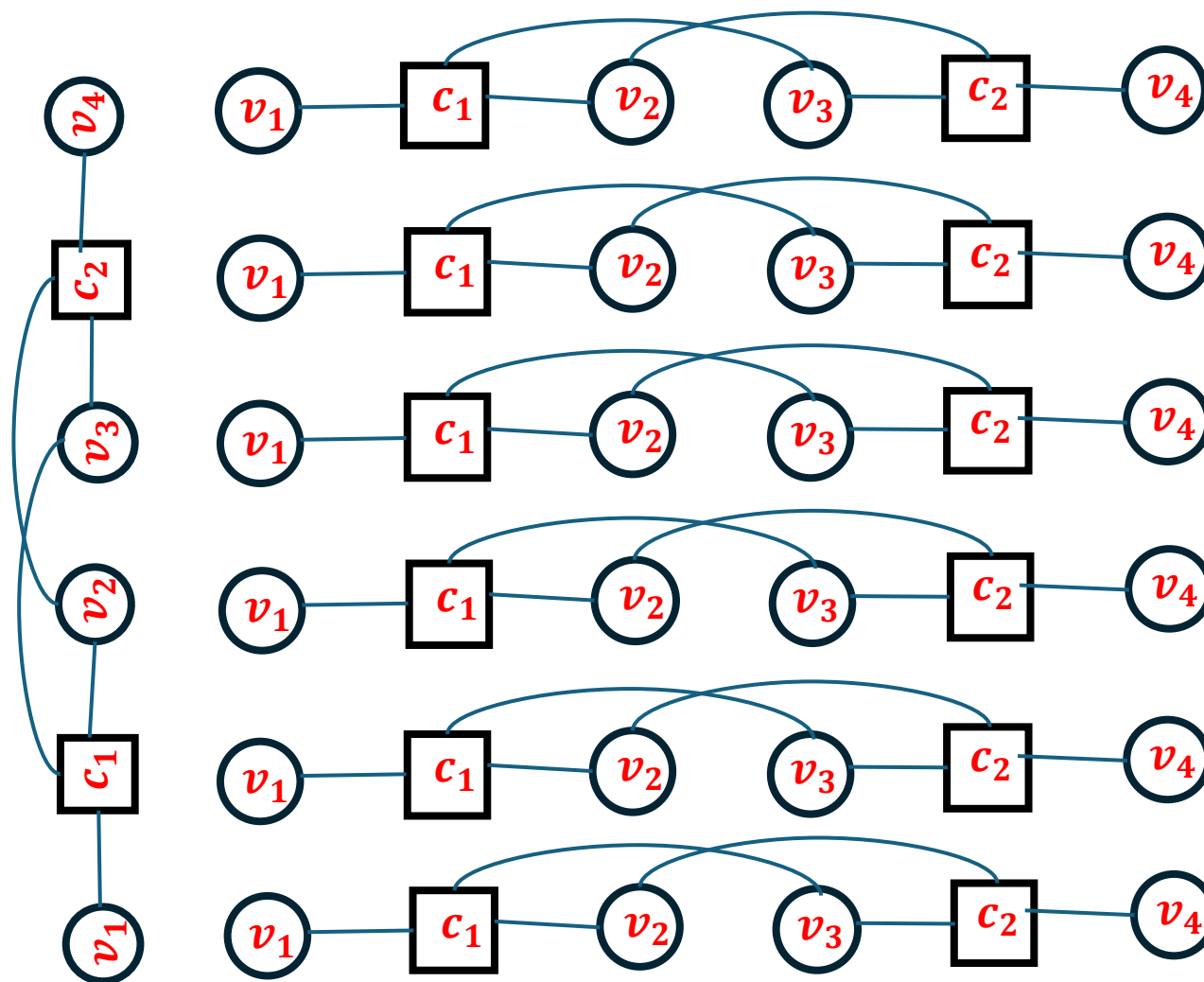
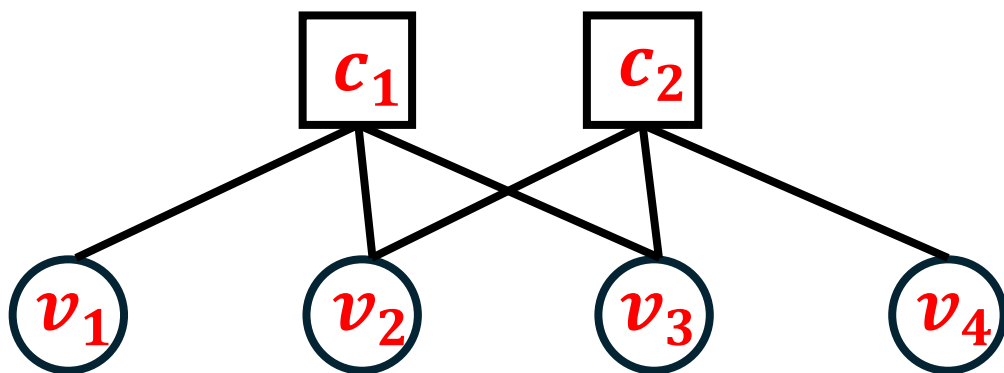


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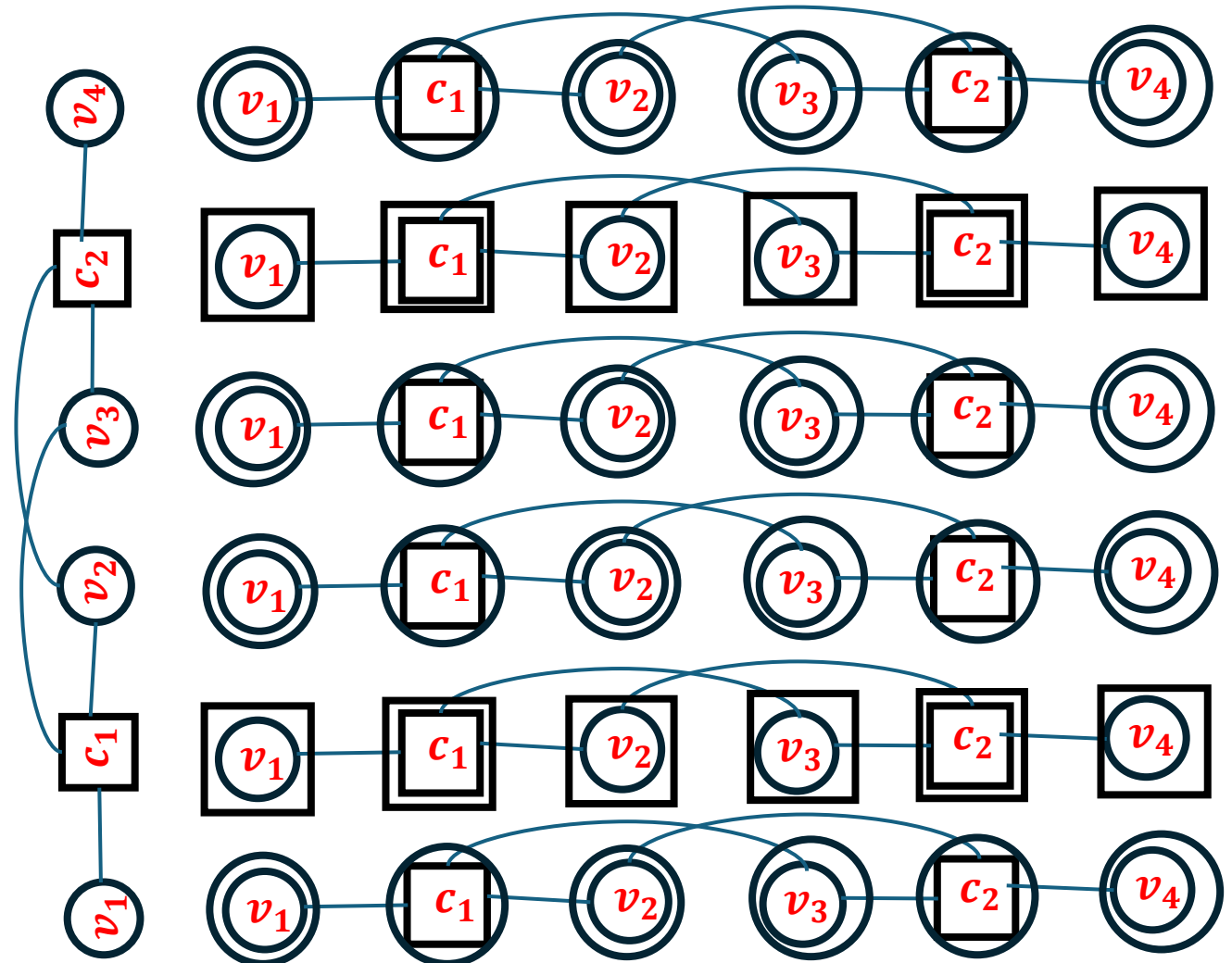
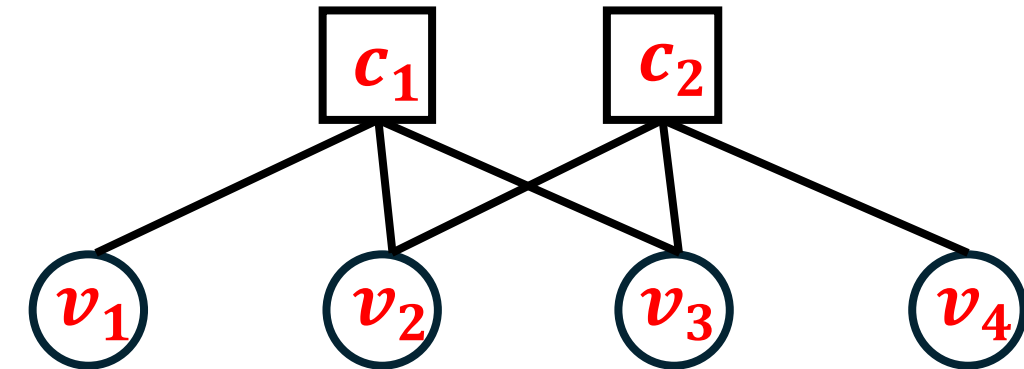


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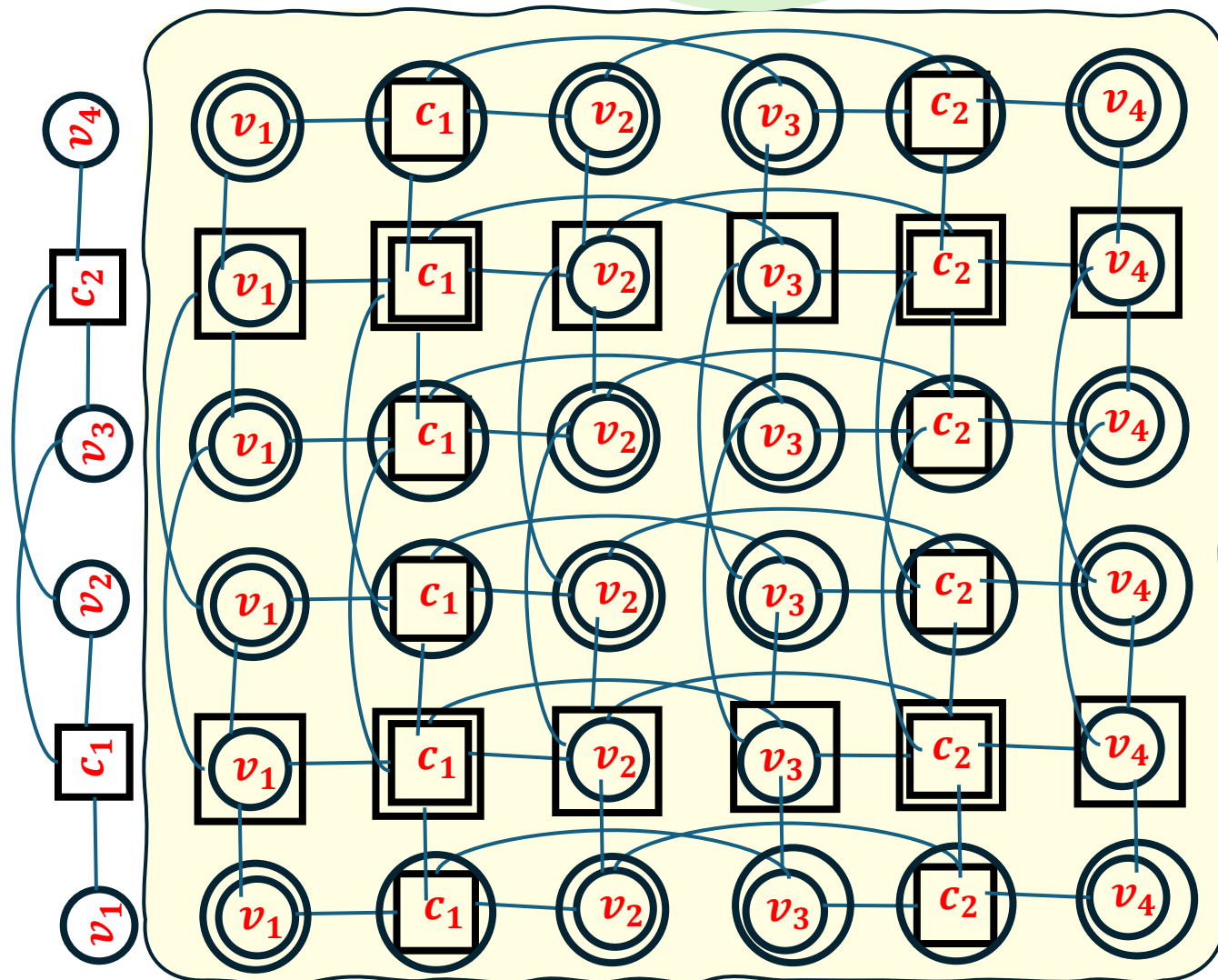
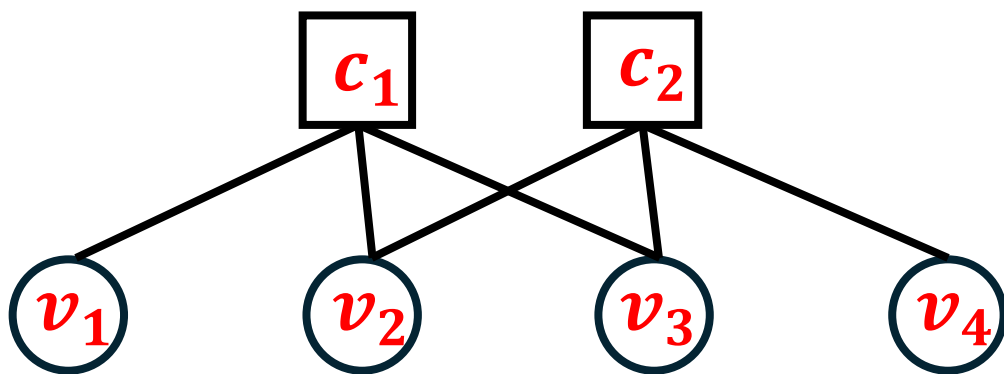


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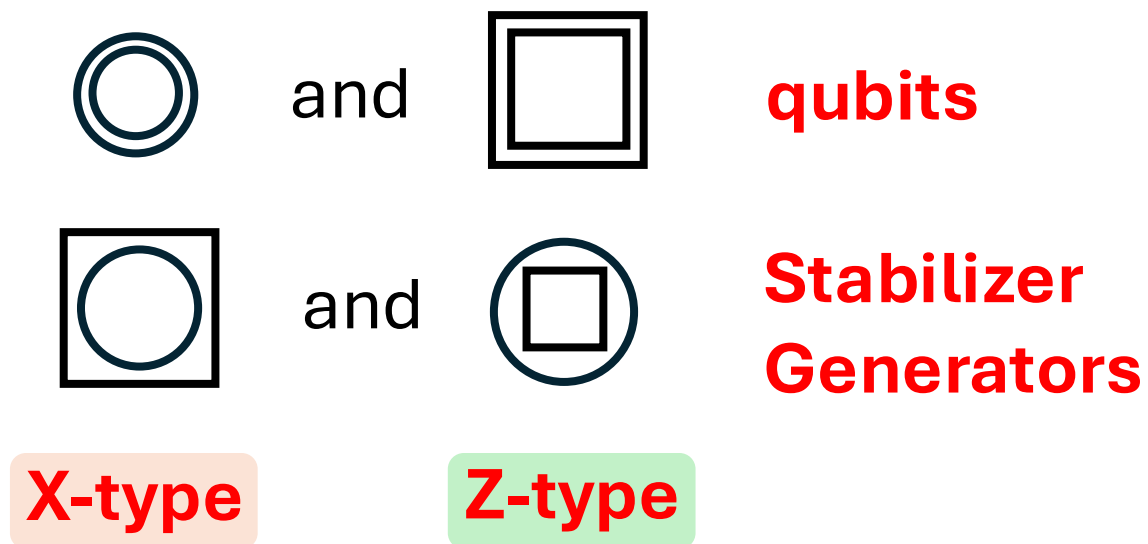
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Hypergraph Product Codes

- Hypergraph product (quantum) code: $(n, k) \times (n, k) \mapsto [[n_1 n_2, k_1 k_2]]$



$$H_Q = \begin{bmatrix} H \otimes I_{n \times n} & I_{m \times m} \otimes H^T \\ I_{n \times n} \otimes H & H^T \otimes I_{m \times m} \end{bmatrix}$$

$H : (m \times n)$ parity check matrix

physical qubits: $n^2 + m^2$, # logical qubits: k^2

